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PROFESSIONAL SUMMARY

High-achieving Computer Engineering graduate of the GUC with international research experience at **Universität Augsburg**. Specializes in **Applied Machine Learning**, **Algorithmic Optimization**, and **Full Stack Development**. Proven track record of academic excellence (Category “A” Scholar) with peer-reviewed research in **wearable-based affective computing** and **multi-agent systems optimization**.

EDUCATION

- **B.Sc. in Computer Engineering** (Media Engineering & Technology) | **German University in Cairo (GUC)** | 2021 – 2026
 - **Bachelor Thesis:** “Integration of Wearable Sensors into Breathing-based Emotion Regulation Application Interface” – Completed during a semester exchange at **Universität Augsburg, Germany (Grade: A+)**.
 - **Cumulative GPA:** 2.09 (German Scale).
 - **Awards:** Category “A” Student (25% Scholarship); 1st place, GUC × RSA Cybersecurity CTF (2026).
- **British High School Diploma (IGCSE)** | **Al-Bashaer International School** | **Grade: 121.25%** | *Graduated 2021*

RESEARCH & WORK EXPERIENCE

- **Wearable-Based Real-Time Stress Detection and Breathing Intervention** | *EAI PervasiveHealth 2025 (in press)*
 - **Lead author.** Built an end-to-end ECG-based stress detection system using a **Random Forest** classifier (**81%** LOSO accuracy on WESAD) integrated with a non-intrusive guided breathing intervention via **React Native / Flask**.
- **Optimization of Multi-AGV Scheduling for Airport Baggage Handling** | *ICEENG-IPCS 2026 (in press)*
 - **Co-author.** Benchmarked SA, GA, ACO, Q-Learning, and a proposed **Hybrid ABC** algorithm on a **VRPTW** formulation; ABC achieved **61%** efficiency gain over FCFS baseline with **40%** fleet reduction.
- **IT Networks and Applications Intern** | National Automotive Company (NATCO) | *June 2024 – Aug 2024*
 - Developed an Employee Records Management System using **C# .NET** and **EF Core**, implementing full CRUD operations.
 - Designed normalized SQL schemas and utilized **LINQ** queries to optimize data retrieval and reporting performance.

PROJECTS

- **ClusterChaos: Amazon Microservices Replica (Team of 13)** | *Java, Spring Boot, Kubernetes, RabbitMQ*
 - Built a 5-service e-commerce backend with per-service polyglot persistence (**PostgreSQL, Cassandra, Neo4j, Elasticsearch, Redis**), a **choreography saga** over RabbitMQ with DLQs and idempotent consumers, a JWT-validating API gateway, and full **K8s** deployment with Prometheus, Grafana, and Loki observability.
- **SBOM Generator (Team of 4)** | *Python, Docker, Click*
 - Built a CLI tool that scans Python/Java projects, resolves transitive dependency trees via **PyPI/Maven Central** APIs, generates schema-validated **CycloneDX 1.7** and **SPDX 3.0** SBOMs, and produces **OSV**-based CVE reports sorted by severity.
- **Intelligent Airline Assistant & Knowledge Graph** | *Python, Neo4j, Scikit-Learn*
 - **Predictive Modeling:** Developed a pipeline to predict passenger satisfaction using Random Forest/FFNN with SHAP/LIME explainability.
 - **Knowledge Graph RAG:** Architected a Neo4j RAG system to answer complex travel queries using vector embeddings and graph traversal.
- **AI Game Agents & Logic Reasoning** | *Java, Prolog*
 - Engineered an autonomous Minimax agent with Alpha-Beta Pruning for battle strategy and developed a Prolog planning agent using Situation Calculus to solve complex logic problems.
- **Custom Database Management System** | *Java*
 - Engineered a high-performance database engine from scratch, implementing B+ Tree indexing, page serialization, and binary search algorithms to optimize CRUD operations.
- **Virtual Trip Planner** | *MERN Stack (MongoDB, Express, React, Node.js)*
 - Architected a full-stack web application for travel budgeting and itinerary planning, integrating third-party APIs for real-time booking data and secure user authentication.

VOLUNTARY WORK & ACTIVITIES

- **Competitive Swimmer:** 15+ years at national level; multiple national and regional medals alongside full-time studies.

SKILLS

- **Programming:** Java, Python, C, SQL, Haskell, Prolog, TypeScript.
- **Frameworks & Tools:** React Native, Node.js, Express.js, Flask, Spring Boot, .NET, Entity Framework, REST APIs, Scikit-learn, Docker, Kubernetes, RabbitMQ, PostgreSQL, MongoDB, Neo4j, JUnit, Git/GitHub, LaTeX.
- **Languages:** Arabic (Native), English (Fluent), German (B1/Intermediate).